

# PETER A. SAYEGH

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## EDUCATION

### Columbia University

New York, NY

#### MS in Electrical Engineering – Nikola Tesla Scholarship, GPA: 3.80/4.0

Expected Dec 2026

Coursework: Digital Signal Processing, Adv. Analog Integrated Circuits, Power Electronics, MOS Transistors, Analog-Digital Interfaces in VLSI, Embedded AI, Microwave Circuit Design, Motor Drive Systems, Nonlinear and Adaptive Control

Activities: Lead Department Representative for Electrical Engineering within the Engineering Student Graduate Council (EGSC)

### École Polytechnique

Palaiseau, FR

#### BS in Mathematics and Physics, GPA: 3.75/4.0

Jun 2025

Coursework: Convex Optimization, Stochastic Processes, Electrodynamics, Adv. Quantum Physics, Thermodynamics, Statistical Physics, Condensed Matter, Algorithms, Numerical Analysis, Web Programming, Object-Oriented Programming, Machine Learning

## EXPERIENCE

### Illumination Diagnostics

New York, NY

#### Electrical Engineering Intern

May 2026 - Present

- Developed a heart rate extraction pipeline for a 12-channel optical wearable, achieving 1-3 bpm clinical accuracy
- Prototyped PPG and temperature acquisition systems using ESP32 microcontrollers with synchronized respiration belt recording
- Assembled an IR LED illumination array on perfboard to provide stable background illumination, reducing ambient light interference
- Designed a 2-axis pan/tilt laser targeting system (dual servos, Arduino + Waveshare driver), achieving  $<0.1^\circ$  angular resolution for programmable wall-target positioning

### TriSpan LLP

New York, NY

#### Private Equity Intern

Jul - Aug 2024

- Streamlined Profit & Loss reporting across 10 portfolio companies using Excel
- Conducted Leveraged Buyout modeling using Excel and Bloomberg terminals
- Drafted valuation reports using Discounted Cash Flow analysis for portfolio companies worth \$200M+ in assets

## ELECTRICAL ENGINEERING PROJECTS

### Columbia University

New York, NY

#### Fully Differential Switched-Capacitor Amplifier

Feb – May 2026

- Designed folded-cascode OTA with CMFB in  $0.18\ \mu\text{m}$  CMOS, achieving 56.6 dB gain,  $71^\circ$  phase margin, and 333 dB CMRR at  $486\ \mu\text{W}$
- Derived settling and slew-rate constraints to meet 0.09% settling error within 28 ns for a 7-bit 16 MHz pipelined ADC

### Columbia University

New York, NY

#### SignSpeak: ASL Recognition on Microcontroller

Feb – May 2026

- Deployed 16K-parameter depthwise-separable CNN on ESP32S3 via TFLite Micro INT8 quantization, achieving 97.4% accuracy in 67.4 KB flash
- Built a full on-device inference pipeline from camera capture to OLED display,  $<110\ \text{ms}$  latency with no cloud dependency

### Columbia University

New York, NY

#### 250 MS/s 8-bit Current-Steering DAC

Feb - May 2026

- Designed a 3+5 segmented current-steering DAC in  $0.18\ \mu\text{m}$  CMOS with thermometer MSBs and per-cell trim DACs, achieving SFDR  $>66\ \text{dB}$  and DNL  $<2 \times 10^{-4}$  LSB across all process corners
- Optimized cascode unit cell sizing to meet 1% mismatch budget, delivering  $\geq 1.53\ V_{\text{pp,diff}}$  into  $100\ \Omega$  at 250 MS/s within 27.6 mW

### École Polytechnique - Center for Theoretical Physics

Palaiseau, FR

#### Dynamic Stability and Levitation Control in Magnetic Levitation (Maglev) Systems

Feb 2024 - Jul 2025

- Designed and constructed a scaled Maglev train prototype using magnetic field analysis, achieving 96% theoretical validation
- Extrapolated results to a real Electromagnetic Suspension engine scale, predicting capacity within 2% of Shanghai Maglev

## TECHNICAL SKILLS

**CAD Tools:** Cadence Spectre, PLECS, LTSpice, Keysight ADS, Xschem, ngspice, Magic VLSI, KLayout, OpenROAD

**Programming Languages:** Python, C++, Arduino, R, MATLAB, HTML, CSS, JavaScript, QML

**Data Science and Machine Learning:** TensorFlow, Keras, JAX/Optax, scikit-learn, NumPy, SciPy, pandas, statsmodels, DSPy

**Scientific Computing:** QuTiP, Scikit-HEP, PySpice

**Languages:** French (Native), Arabic (Native), German (Advanced), Spanish (Intermediate)